EE4-24
EE9-CS1-3

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

MSc and EEE/EIE PART IV: MEng and ACGI

Corrected Copy

DISCRETE-TIME SYSTEMS AND COMPUTER CONTROL

Tuesday, 21 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : G. Scarciotti
Second Marker(s) : A. Astolfi

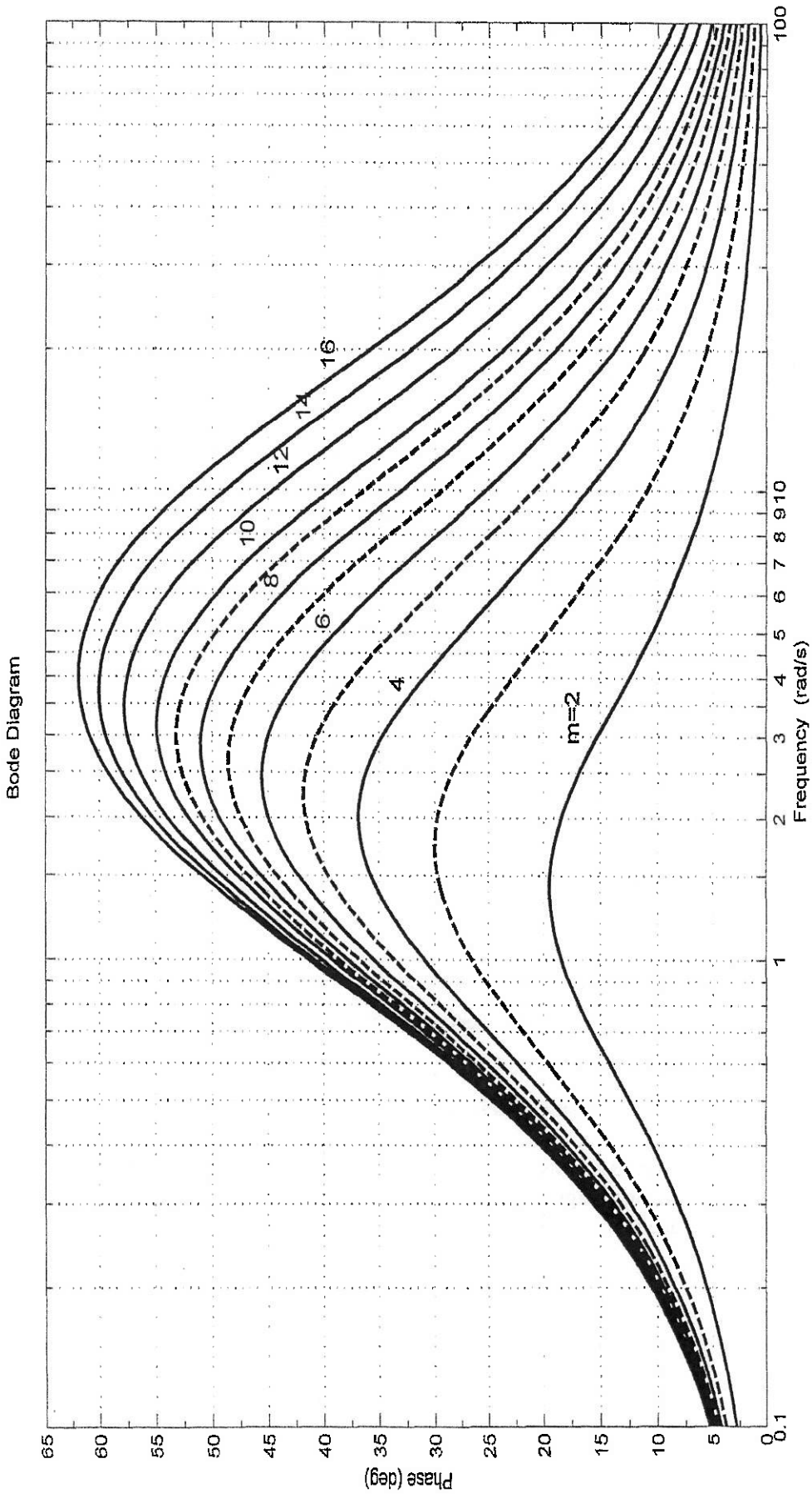
© Imperial College London

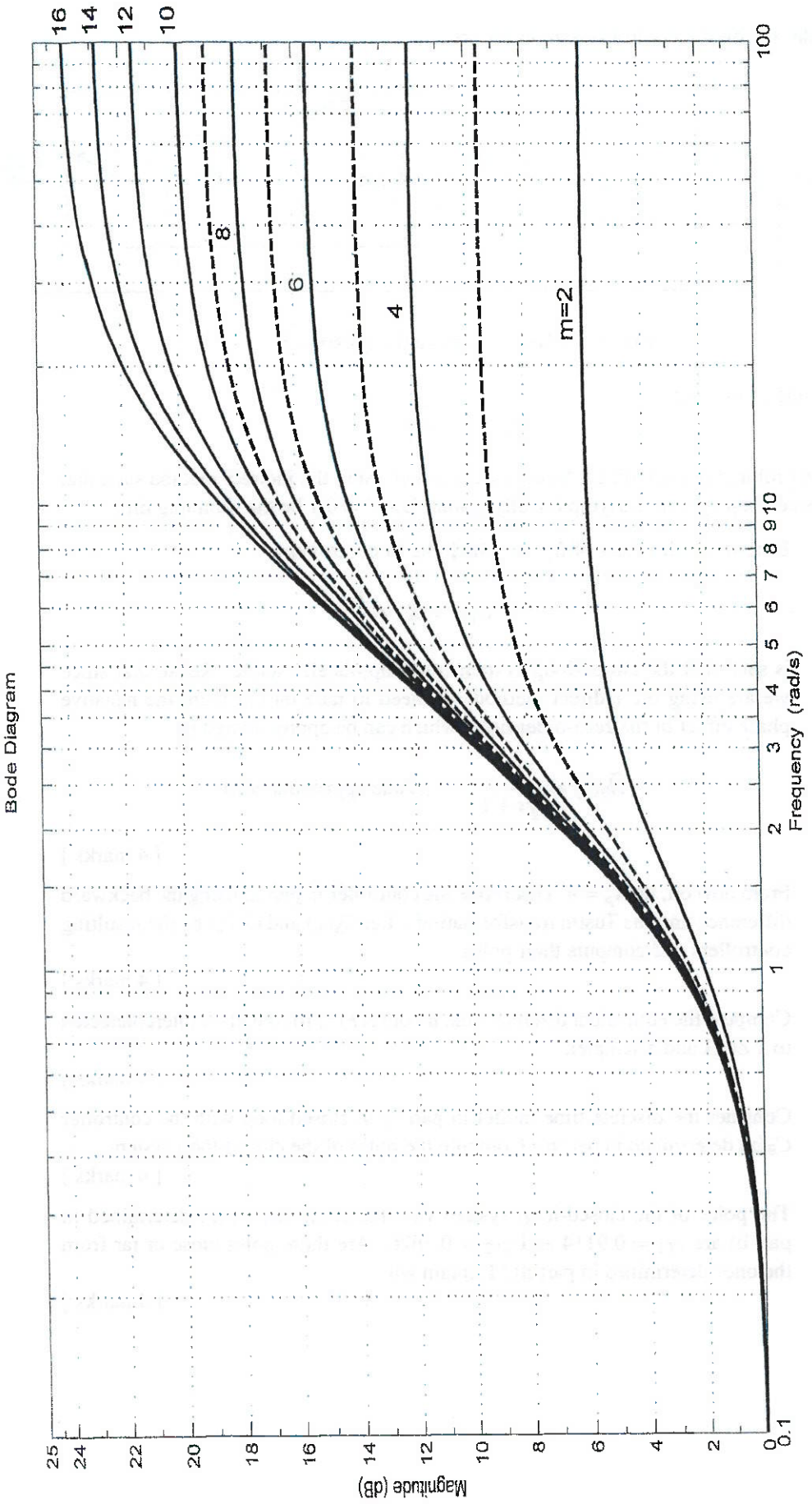
DISCRETE-TIME SYSTEMS AND COMPUTER CONTROL

Information for candidates:

- $Z\left(\frac{1}{s}\right) = Z(1(k)) = \frac{z}{z-1}$
- $Z\left(\frac{1}{s+a}\right) = Z\left(e^{-akT}\right) = \frac{z}{z-e^{-aT}}$
- $Z\left(\frac{b}{(s+a)^2+b^2}\right) = Z\left(e^{-akT}\sin bkT\right) = \frac{ze^{-aT}\sin bT}{z^2-2ze^{-aT}\cos bT+e^{-2aT}}$
- $Z\left(\frac{s+a}{(s+a)^2+b^2}\right) = Z\left(e^{-akT}\cos bkT\right) = \frac{z^2-ze^{-aT}\cos bT}{z^2-2ze^{-aT}\cos bT+e^{-2aT}}$
- $Z\left(\frac{1}{s^2}\right) = Z(kT) = \frac{zT}{(z-1)^2}$
- $Z\left(\frac{1}{s^3}\right) = Z\left(\frac{1}{2}(kT)^2\right) = \frac{T^2z(z+1)}{2(z-1)^3}$
- Transfer function of the ZOH: $H_0(s) = \frac{1-e^{-sT}}{s}$
- Definition of the w-plane: $z = \frac{1+\frac{wT}{2}}{1-\frac{wT}{2}}$, $w = \frac{2}{T}\frac{z-1}{z+1}$
- Tustin transformation: $s = \frac{2}{T}\frac{z-1}{z+1}$
- Note that, for a given signal r or $r(t)$, $R(z)$ denotes its Z-transform. All signals are assumed to be zero for negative times.
- Let $C(s) = \frac{1+\tau s}{1+\frac{\tau}{m}s}$ be a compensator parametrized by a positive integer m and a positive real number τ. The Bode diagrams of this compensator for $\tau = 1$ and different values of m are shown in the next two pages. The first page shows the phase plot parametrized by m, whereas the second page shows the magnitude plot parametrized by m. Recall that if $\hat{\omega}$ is the frequency at which you want to place the phase peak of the compensator, then $\tau = \frac{\sqrt{m}}{\hat{\omega}}$.

 ©Imperial College London


Bode Diagram
Phase (deg)
Frequency (rad/s)
m=2
4
6
8
10
12
14
16
65
60
55
50
45
40
35
30
25
20
15
10
5
0
0.1
1
2
3
4
5
6
7
8
9
10
100


Bode Diagram
Magnitude (dB)
Frequency (rad/s)
m=2
4
6
8
10
12
14
16

1. Consider the digital control system in Figure 1.1

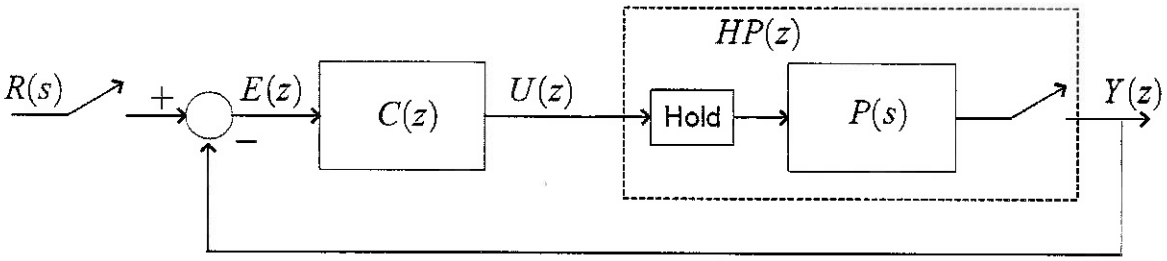


Figure 1.1 Block diagram for question 1.

with transfer function

$$P(s) = \frac{1}{s-1}$$

and the problem of designing a feedback controller using the indirect method such that the closed-loop system is asymptotically stable. Let $T = 0.1$ be the sampling time.

a) Determine all values of K_p for which the PI control law

$$C(s) = K_p + \frac{2}{s}$$

is such that the closed-loop system is asymptotically stable. Recall that since we are using the indirect method, we need to take into account the negative phase effect of the zero-order hold, which can be approximated by

$$\tilde{H}_0(s) \approx \frac{1}{\frac{T}{2}s+1}. \qquad \text{(Padé approximation)}$$

[4 marks]

b) From now on, let $K_p = 4$. Discretize the controller in part a) using the backward difference and the Tustin transformations. Let $C_B(z)$ and $C_T(z)$ be the resulting controllers and compute their poles.

[4 marks]

c) Compute the equivalent discrete-time model $HP(z)$ for the plant interconnected to a ZOH and a sampler.

[4 marks]

d) Consider the discrete-time model in part c) in closed-loop with the controller $C_B(z)$ determined in part b). Compute the poles of the closed-loop system.

[4 marks]

e) The poles of the closed-loop system with the controller $C_T(z)$ determined in part b) are $z_{T1} = 0.9114$ and $z_{T2} = 0.7626$. Are these poles close or far from the ones determined in part d)? Explain why.

[4 marks]

2. Consider the digital control system in Figure 2.1.

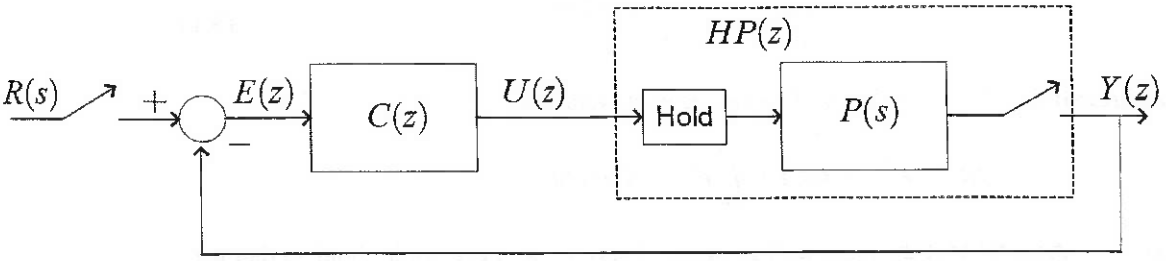


Figure 2.1 Block diagram for question 2.

The discrete-time behaviour of the plant (*i.e.* $HP(z)$) is described by the difference equation

$$y(k)-2y(k-1)+y(k-2)=2u(k-1)+u(k-2).$$

a) Determine the transfer function $\dfrac{Y(z)}{U(z)}$.

[2 marks]

b) Consider a controller described by the transfer function

$$C(z)=\frac{s_0z+s_1}{z+v_1}$$

and the problem of designing a deadbeat controller (all closed-loop poles are at zero). Justify why the problem has a unique solution and determine the values of s_0, s_1 and v_1 that solve the problem.

[6 marks]

c) Compute the closed-loop transfer function $\dfrac{Y(z)}{R(z)}$.

[3 marks]

d) Compute the output sequence $y(k)$ for $k \geq 0$ when the reference signal is $r(k) = 1$ for $k \geq 0$.

[6 marks]

e) Without further computation, explain how you expect $e(k) = r(k) - y(k)$ to behave for large k when the reference signal is:

i) $r(k)=k$;

ii) $r(k)=k^2$;

iii) $r(k)=k^3$.

[3 marks]

3. Consider the continuous-time system

$$\begin{aligned} \dot{x}(t) &= Fx(t)+Gu(t), \\ y(t) &= Hx(t), \end{aligned} \tag{3.1}$$

and recall that the state $x(t)$ of (3.1) can be expressed as

$$x(t) = e^{F(t-t_0)}x(t_0) + \int_{t_0}^{t} e^{F(t-\tau)}Gu(\tau)d\tau \qquad t \geq t_0.$$

a) Assume that the input $u(t)$ is sampled and fed to a zero-order hold so that $u(t)$ is constant over the interval between two consecutive sampling instants. Prove that the behaviour of the discretized system is described by

$$\begin{aligned} x(k+1) &= Ax(k)+Bu(k), \\ y(k) &= Cx(k), \end{aligned} \tag{3.2}$$

with

$$A = e^{FT}, \qquad B = \int_0^T e^{F\lambda}d\lambda G, \qquad C = H.$$

Hint: pick $t = (k+1)T$ and $t_0 = kT$. Then use the change of variable $\lambda = (k+1)T - \tau$.

[4 marks]

b) Let

$$F = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \qquad G = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad H = \begin{bmatrix} 0 & 1 \end{bmatrix}.$$

Compute the matrices A and B defined in part a).
Hint: recall that $e^{Ft} = \mathscr{L}^{-1}\left[(sI-F)^{-1}\right]$, where $\mathscr{L}^{-1}$ indicates the inverse Laplace transform operator.

[4 marks]

c) Study the reachability properties of the pair (A,B) and determine values of T such that (A,B) is unreachable.

[4 marks]

d) Study the observability properties of the pair (A,C) and determine values of T such that (A,C) is unobservable.

[4 marks]

e) Compute the input-output transfer function of the discrete-time system (3.2). For which values of T the transfer function is identically zero and why?

[4 marks]

4. Consider a process to be controlled with transfer function

$$P(s) = \frac{1}{s^2}.$$

Assume the system is interconnected to a ZOH and a sampler. Let $T = 0.1$ be the sampling time.

a) Compute the equivalent discrete-time model $HP(z)$ for the plant interconnected to the hold and the sampler.

[4 marks]

b) Using the definition of the w-plane show that the transfer function $HP(w)$ is

$$HP(w) = \frac{1 - 0.05w}{w^2}.$$

[3 marks]

c) Let

$$C_1(w) = K,$$

Select K such that the system has acceleration constant $K_a = 4$. Determine the open-loop transfer function obtained from the interconnection of $C_1(w)$ with the transfer function $HP(w)$.
Hint: recall that, in the s-plane, the acceleration constant of a function $G(s)$ is given by $K_a = \lim_{s\to 0} s^2\, G(s)$.

[3 marks]

d) Let

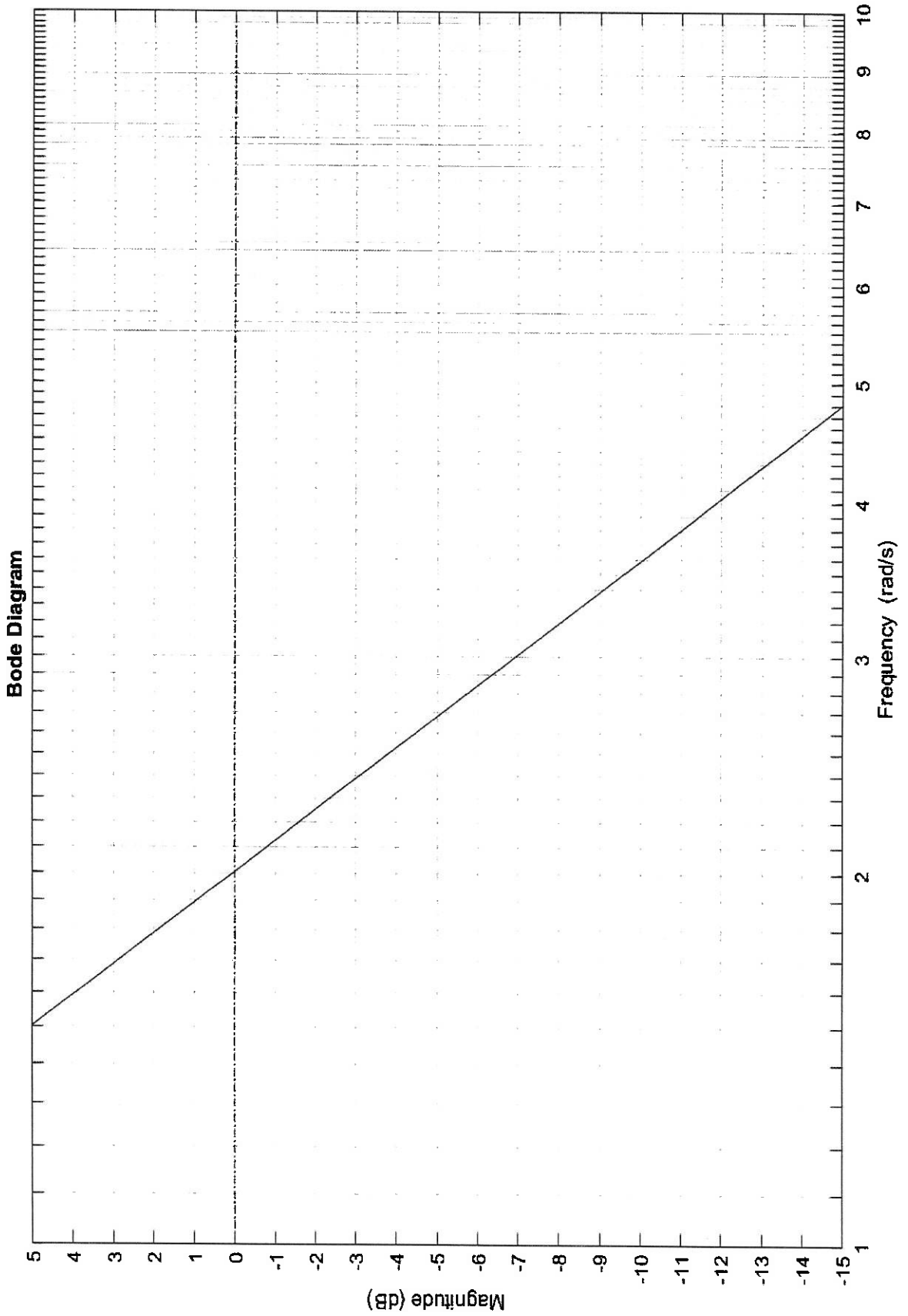
$$C_2(w) = \frac{1 + \tau w}{1 + \frac{\tau}{m} w},$$

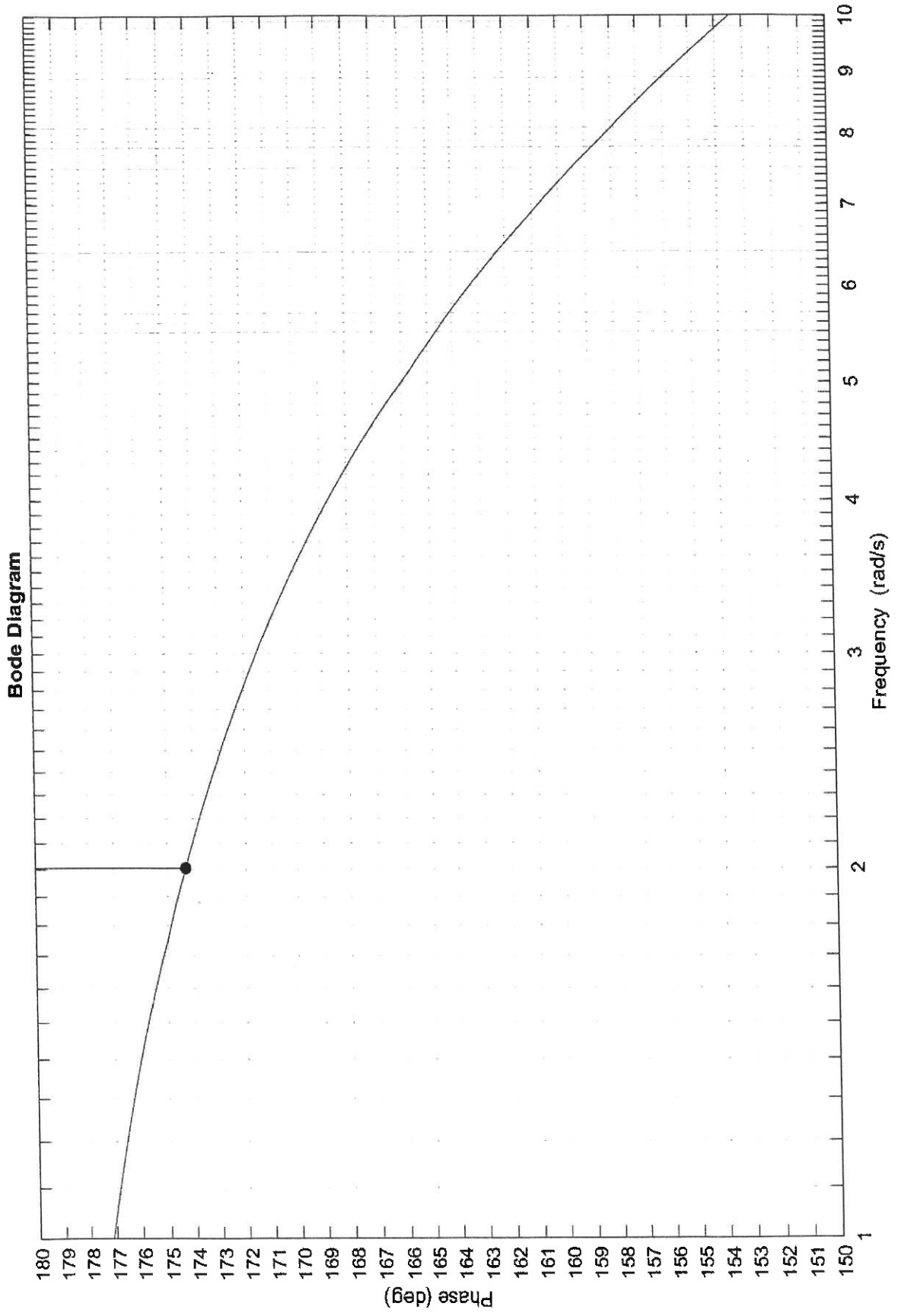
with $\tau > 0$ and $m > 0$ to be determined. Consider the closed-loop system resulting from the unity feedback interconnection of the controller $C_2(w)$ with the transfer function $C_1(w)HP(w)$ obtained in part c). Determine the controller $C_2(w)$ such that the phase margin is at least 50°. To facilitate the design, the Bode plot of the correct $C_1(w)HP(w)$ is given in the next two pages. The Bode plot of the phase-lead compensator is given at the beginning of this exam paper.

[6 marks]

e) Using the values determined in part c) and d), compute the discrete-time controller $C_1(z)C_2(z)$.

[4 marks]


Bode Diagram
Magnitude (dB)
Frequency (rad/s)
5
4
3
2
1
0
-1
-2
-3
-4
-5
-6
-7
-8
-9
-10
-11
-12
-13
-14
-15
1
2
3
4
5
6
7
8
9
10


Bode Diagram
Phase (deg)
Frequency (rad/s)
180
179
178
177
176
175
174
173
172
171
170
169
168
167
166
165
164
163
162
161
160
159
158
157
156
155
154
153
152
151
150
1
2
3
4
5
6
7
8
9
10